BCSC 0815: DATA VISUALIZATION LAB

Objective: *The objective of this course is that students will understand and implement all kind of graph and plots for visualizing and analyzing different kind of data to attain the business objective.*

Credits: 01 **L–T–P-J: 0–0–2-0**

Module No.	Content	Hours
I/II	**1. Basic Data Exploration with numpy, Pandas and Matplotlib** **2. Missing Values Treatment: Handling Missing Data** **3. Understanding the Data: Data Profiling and Initial Insights** **4. Univariate Analysis: Univariate Analysis with Visualizations** **5. Bivariate Analysis: Exploring Relationships Between Two Variables** **6. Outliers Treatment: Detecting and Handling Outliers** **7. Experiment on different types of Plots** **a. Line Plots** **b. Area Plots** **c. Histograms** **d. Bar Charts** **e. Pie Charts** **f. Box Charts** **g. Scatter Plots** **8. Introduction to Plotly: Creating Basic Plots Using Plotly** **9. Introduction to Dash: Creating a Basic Dashboard Using Dash** **10. Interactive Dashboard with Multiple Plots: Combining Multiple Plotly Graphs in One Dashboard**	24

Outcome: After completion of course, student will be able to:

- CO1: Demonstrate the plot for continuous data.
- CO2: Demonstrate the univariate and bivariate plot for categorical data.
- CO3: Demonstrate the univariate and bivariate plot for time series data.